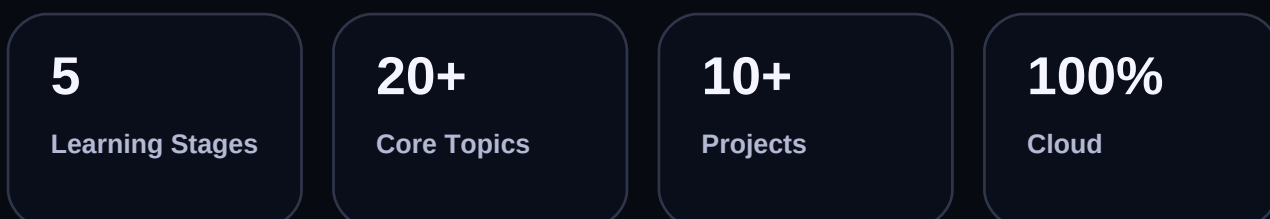
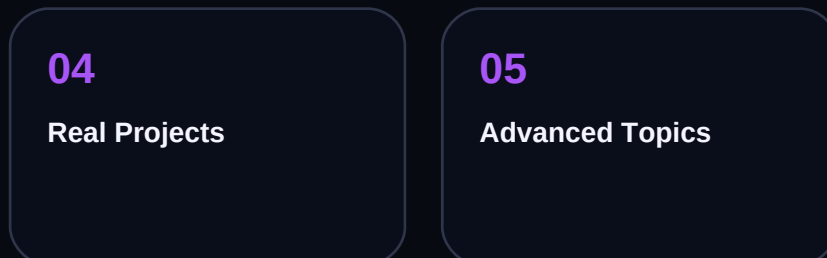
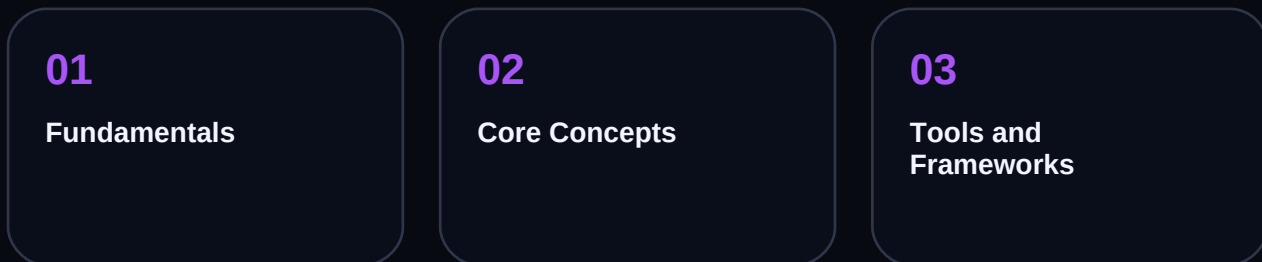
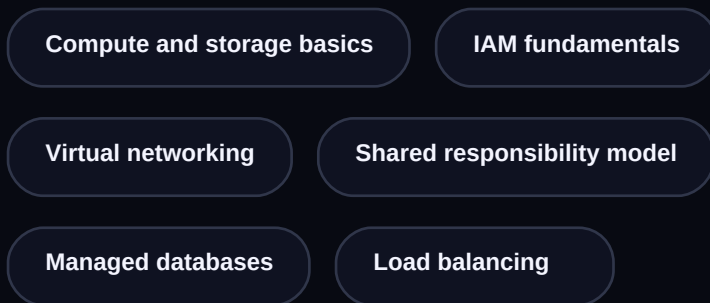




CLOUD ENGINEERING

Complete Learning Roadmap

Design, deploy, and operate cloud-native systems with reliability and cost discipline.



Learning Path Overview

Five progressive stages from fundamentals to production mastery

01

Fundamentals

Compute and storage basics • IAM fundamentals • Virtual networking • Shared responsibility model

02

Core Concepts

Managed databases • Load balancing • Autoscaling • Cloud logging and metrics

03

Tools and Frameworks

AWS/Azure/GCP core services

04

Real Projects

Multi-tier app deployment • Global CDN setup • Backup strategy implementation • Cost-aware environments

05

Advanced Topics

Multi-region resilience • Policy-as-code • Cloud governance • Security posture automation

How to Use This Roadmap

Build and ship first

Complete a small working project in every stage before advancing.

Use official sources

Start with the docs and use secondary resources to reinforce implementation patterns.

Iterate on projects

Refactor earlier work as you learn better architecture, testing, and performance habits.

Stages 01 & 02 - Deep Dive

Fundamentals, core concepts, and architectural foundations

01

Stage 1 - Fundamentals

Estimated: 3-4 weeks | Prerequisites: Basic foundations

Topics

- Compute and storage basics
- IAM fundamentals
- Virtual networking
- Shared responsibility model

Tools & Frameworks

- Compute and storage basics
- IAM fundamentals
- Virtual networking
- Shared responsibility model

Recommended Projects

- Cloud Engineering workshop using Compute and storage basics
- Cloud Engineering portfolio project with IAM fundamentals

Stage 1 Milestone Complete Cloud Engineering workshop using Compute and storage basics and document the decisions, trade-offs, and validation checks before moving to Stage 2.

02

Stage 2 - Core Concepts

Estimated: 4-5 weeks | Prerequisites: Stage 1 complete

Topics

- Managed databases
- Load balancing
- Autoscaling
- Cloud logging and metrics

Tools & Frameworks

- Managed databases
- Load balancing
- Autoscaling
- Cloud logging and metrics

Recommended Projects

- Cloud Engineering workshop using Managed databases
- Cloud Engineering portfolio project with Load balancing

Stage 2 Milestone Complete Cloud Engineering workshop using Managed databases and document the decisions, trade-offs, and validation checks before moving to Stage 3.

Stages 03 & 04 - Deep Dive

Modern tooling, delivery frameworks, and portfolio projects

03

Stage 3 - Tools and Frameworks

Estimated: 4-6 weeks | Prerequisites: Stage 2 complete

Topics

- AWS/Azure/GCP core services
- Serverless tooling
- Container platforms
- Cost-management tools

Tools & Frameworks

- AWS/Azure/GCP core services

Recommended Projects

- Cloud Engineering workshop using AWS/Azure/GCP core services
- Cloud Engineering portfolio project with Serverless tooling

Stage 3 Milestone Complete Cloud Engineering workshop using AWS/Azure/GCP core services and document the decisions, trade-offs, and validation checks before moving to Stage 4.

04

Stage 4 - Real Projects

Estimated: 6-8 weeks | Prerequisites: Stage 3 complete

Topics

- Multi-tier app deployment
- Global CDN setup
- Backup strategy implementation
- Cost-aware environments

Tools & Frameworks

- Multi-tier app deployment
- Global CDN setup
- Backup strategy implementation
- Cost-aware environments

Recommended Projects

- Cloud Engineering workshop using Multi-tier app deployment
- Cloud Engineering portfolio project with Global CDN setup

Stage 4 Milestone Complete Cloud Engineering workshop using Multi-tier app deployment and document the decisions, trade-offs, and validation checks before moving to Stage 5.



Stage 05 - Advanced Topics & Delivery

Performance, architecture, release automation, and production governance

05

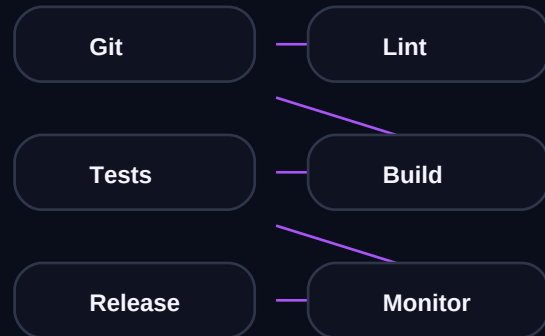
Stage 5 - Advanced Topics

Estimated: Ongoing | Industry-grade operating habits

Advanced Focus

- Multi-region resilience
- Policy-as-code
- Cloud governance
- Security posture automation

CI/CD Pipeline



Stage 5 Milestone

A production-grade cloud engineering workflow with measurable performance, quality, and release confidence.

Performance Budget

Set and track concrete engineering thresholds: startup latency, error budget, throughput, cost efficiency, and deployment confidence.

Resources & Practice Questions

Verified links, self-assessment, and community follow-up

FREE

Cloud Engineering docs and community guides

Link: <https://docs.aws.amazon.com/>

FREE

Build one portfolio project per stage before moving ahead

Link: <https://roadmap.sh/aws>

PAID

Structured specialization track focused on Cloud Engineering

Link: <https://roadmap.sh/aws>

FREE

Cloud Engineering roadmap reference

Link: <https://roadmap.sh/aws>

FREE

Official documentation

Link: <https://docs.aws.amazon.com/>

Self-Assessment Practice Questions

- How would you evaluate readiness for Stage 3 in Cloud Engineering?
- Which project best proves Stage 4 competency in Cloud Engineering?
- What trade-offs appear in Stage 5 decisions for Cloud Engineering?